IndiaGround: An Automated Dual-Pipeline System for News Bias Detection and Fact-Checking in the Indian English Media Ecosystem

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ABSTRACT

India operates one of the world's largest and most complex English-language news ecosystems, yet no open, automated system currently evaluates Indian English news articles for bias and factual reliability in real time. This paper presents IndiaGround, a production-grade platform that applies a dual-pipeline natural-language-processing architecture to this problem. The architecture orthogonally separates how a story is written — stylistic bias, framing and loaded language, processed by the unBIAS module — from what it claims — factual verifiability and evidentiary support, handled by the ClaimBuster module. Both pipelines feed a weighted aggregator that produces a decomposed Reliability Score on a [0, 100] scale. The system incorporates India-specific adaptations including a curated lexicon of over ninety politically loaded terms with neutral alternatives, six India-specific zero-shot framing labels covering communal/divisive and nationalistic framing absent from Western schemas, and a credibility database covering more than one hundred Indian and international news sources. The system is deployed as a FastAPI backend with eleven REST endpoints, a React 19 frontend with seven pages, a Celery/Redis asynchronous task queue, a PostgreSQL 18 store and a Docker Compose orchestration. An evaluation framework benchmarks the pipeline against BABE, LIAR and CLEF CheckThat!. To the best of our knowledge, IndiaGround is the first open-source, deployable system that combines automated bias detection and fact-checking specifically calibrated for the Indian English news ecosystem.

KEYWORDS: *news bias detection, automated fact-checking, Indian media, natural language processing, zero-shot classification, BART-MNLI, RoBERTa, story clustering, media reliability, framing analysis.*

1 INTRODUCTION

India's news media landscape is among the most expansive in the world, spanning over nine hundred satellite television channels, tens of thousands of registered newspapers and a rapidly expanding digital news sector whose growth has tracked the country's smartphone adoption curve. This breadth, while democratising in principle, has been accompanied by well-documented challenges: partisan editorial decisions, widespread sensationalism, communal framing during sectarian events and surges of misinformation during elections and public-health crises. The English-language press, which functions as the lingua franca of national political discourse, is not exempt from any of these pathologies.

The computational tools available to study and counter these dynamics remain predominantly designed for Western, largely American, media contexts. Commercial systems such as Ground News, NewsGuard and Media Bias/Fact Check operate with manual editorial curation and cover Indian outlets only superficially. Research systems such as DBIAS[1] focus on English-language media without India-specific calibration; benchmarks such as BABE[2] and NewsFrames[3] are similarly evaluated on Western English corpora. The systematic review by Rodrigo-Gines et al.[4] identifies three dominant paradigms in bias detection — lexicon-based approaches, supervised classification and zero-shot or few-shot methods — but catalogues no systems calibrated for South Asian English political discourse.

IndiaGround addresses this gap. Its principal contributions are: (1) a dual-pipeline architecture that independently scores stylistic bias and factual reliability and combines them into a unified, decomposed Reliability Score; (2) India-specific NLP adaptations including a curated ninety-term bias lexicon with neutral alternatives, six zero-shot framing labels covering communal and nationalistic framing absent from Western schemas, and a credibility database of more than one hundred Indian and international sources; (3) a two-pass zero-shot check-worthiness detection stage based on BART-MNLI[5] combined with DuckDuckGo-based evidence retrieval and NLI verification[6]; (4) cross-source story clustering via all-MiniLM-L6-v2[7] sentence embeddings, enabling per-story bias-spectrum analysis across outlets; and (5) a production-grade, fully open-source deployment with an evaluation framework benchmarked against BABE[2], LIAR[8] and CLEF CheckThat![9].

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the system architecture, scoring methodology and infrastructure. Section 4 walks through an end-to-end system demonstration. Section 5 reports the evaluation methodology and results. Section 6 compares IndiaGround with existing systems. Section 7 discusses limitations and future work. Section 8 concludes.

2 RELATED WORK

2.1 Automated Bias Detection

Automated news-bias detection has attracted substantial scholarly attention. Rodrigo-Gines et al.[4] provide a comprehensive systematic review identifying three dominant paradigms — lexicon-based approaches, supervised classification and zero-shot or few-shot methods enabled by large pre-trained language models. Early lexicon approaches, typified by VADER[10], offer speed and interpretability but lack contextual understanding and struggle with irony and cultural register. The introduction of transformer-based models substantially advanced the field: Liu et al.[11] demonstrated the power of robustly pre-trained RoBERTa for fine-grained text classification, which systems such as DBIAS subsequently exploit.

Raza et al.[1] propose DBIAS, a BERT-based pipeline that detects biased phrases, classifies articles and generates debiased text. Spinde et al.[2] introduce the BABE dataset, comprising 3,700 sentences from 1,700 news articles annotated by eight trained raters with Krippendorff's $\alpha = 0.49$. Its annotation guidelines follow the linguistic-bias taxonomy of Recasens et al.[12]. Hamborg[3] develops the NewsFrames framework focusing on framing effects rather than surface-level sentiment. While these systems represent the state of the art, all are designed and evaluated on Western English media. IndiaGround differs in three respects: it employs zero-shot classification requiring no fine-tuning, it incorporates India-specific lexical and framing resources, and it treats bias detection as one component of a broader reliability-scoring system rather than a standalone task.

2.2 Automated Fact-Checking

Automated fact-checking has been surveyed extensively by Guo et al.[6], who identify a four-stage pipeline common to most systems: claim detection, evidence retrieval, claim verification and verdict prediction. Hassan et al.[13] pioneered end-to-end automated fact-checking with ClaimBuster, combining an SVM-based check-worthiness scorer with search-based evidence retrieval. The FEVER shared task[14] established the Natural Language Inference paradigm for claim verification: given a claim and evidence passages, a model classifies the relationship as SUPPORTS, REFUTES or NOT ENOUGH INFO. This paradigm, using models such as BART-MNLI[5], now dominates automated fact-checking research. The LIAR dataset[8] provides 12,836 short political statements with six-class truthfulness labels from PolitiFact, and the CLEF CheckThat! shared task[9] provides sentence-level check-worthiness annotations from political speeches and debates.

2.3 Commercial Bias-Rating Platforms

Commercial platforms such as Ground News, NewsGuard and AllSides provide bias ratings to general audiences. Ground News aggregates human-curated MBFC[15] ratings to display how multiple outlets cover the same story. However, Ground News is proprietary, relies entirely on manual editorial curation, provides minimal coverage of Indian outlets and performs no automated fact-checking. NewsGuard employs journalists to manually rate outlets on nine credibility criteria, offering no per-article analysis. IndiaGround provides comparable story-clustering and bias-spectrum functionality to Ground News, adds automated per-article fact-checking absent from all three commercial platforms, and does so through open-source, deployable software specifically calibrated for Indian media.

2.4 Benchmark Datasets

Our evaluation draws on four publicly available datasets. The BABE dataset[2] provides 3,700 annotated sentences for sentence-level bias detection. The LIAR dataset[8] provides 12,836 political statements for veracity classification. The CLEF CheckThat! shared task[9] provides claim check-worthiness annotations from political discourse. The MBIC corpus[16] provides token-level and sentence-level bias annotations across 1,700 articles, enabling evaluation of our token-level flagging pipeline. Additionally, the NELA-GT dataset[17] provides source-level reliability labels enabling validation of our source credibility database. All five datasets originate from Western English media; we use them as sanity checks and relative comparison tools rather than as primary optimisation targets, consistent with our zero-shot design philosophy.

3 SYSTEM ARCHITECTURE AND METHODOLOGY

This section presents the complete architecture, scoring methodology and infrastructure for IndiaGround. All pipeline components — ingestion, stylistic bias analysis, factual verification, score aggregation and story clustering — are described in full, with each modelling choice grounded in prior work and each formula tied to an explicit empirical or theoretical rationale.

3.1 Design Philosophy

IndiaGround is organised around a single architectural principle: the quality of a news article has two orthogonal dimensions that must be evaluated independently before being combined. The first, how a story is written, concerns stylistic signals — loaded language, sensationalist framing, sentiment extremity — that distort reader perception regardless of whether the underlying facts are accurate. The second, what a story claims, concerns factual verifiability — whether the specific assertions can be corroborated by independent evidence. Conflating these dimensions, as many prior systems do, produces scores that are difficult to interpret. IndiaGround addresses this by routing every article through two parallel modules, unBIAS and ClaimBuster, whose outputs are combined by a weighted aggregator.

A second design principle is zero-shot operation. No component of IndiaGround is fine-tuned on BABE, LIAR or any other benchmark. This decision is driven by the absence of annotated Indian English bias-detection data and the risk that fine-tuning on Western benchmarks would introduce calibration error for Indian political discourse. Zero-shot classification via BART-MNLI[5] allows domain knowledge to be encoded through hypothesis-text engineering rather than label curation — a more transparent and adjustable mechanism that accommodates the India-specific framing dimensions absent from Western annotation schemas. The acknowledged trade-off is that zero-shot accuracy is generally lower than fine-tuned accuracy on standard benchmarks; we document this gap in Section 5. A third principle is full

decomposability: every component of every score is stored separately in the database and exposed via the API and frontend, so users can inspect each contribution to the final reliability number.

3.2 Data Ingestion and Preprocessing

The system ingests news from three source types. Inshorts provides a JSON API delivering sixty-word English summaries across ten topic categories. Ten direct Indian outlet RSS feeds are scraped covering The Hindu, NDTV, Indian Express, Times of India, Hindustan Times, Scroll.in, The Wire, Republic World, Zee News and Livemint. Google News RSS supplies additional cross-outlet aggregation. Article scraping is triggered by Celery Beat every thirty minutes.

Deduplication is performed at two levels. Exact duplicates are identified via SHA-256 hashing of article content. Near-duplicate semantic content is identified via cosine similarity of all-MiniLM-L6-v2[7] embeddings at a threshold of 0.92, catching articles where different outlets rewrite the same wire copy with minor changes. Each article then passes through a preprocessing stage using spaCy en_core_web_sm for named-entity recognition and tokenisation, and langdetect for language filtering (English only). Articles advance through a status pipeline raw → preprocessed → analyzed → failed. The full analysis pipeline is managed asynchronously by Celery workers with Redis as the message broker.

3.3 The unBIAS Module: Stylistic Bias Detection

The unBIAS module analyses how a news article is written, producing a composite bias score $B \in [0, 1]$ from four sub-components. The question it answers is whether the article's language and framing introduce non-neutral signals independent of the facts it reports.

3.3.1 Sentiment Analysis

Sentiment extremity is measured as the absolute value of a combined score from headline and body text. Two complementary models are used: VADER[10], a lexicon-based model returning a compound score in [−1, 1] well-suited to short, headline-style text, and a RoBERTa-based sentiment classifier[11] (cardiffnlp/twitter-roberta-base-sentiment-latest) returning three-class probabilities normalised to [−1, 1], providing contextual understanding for longer body text. The combined score is

$$sentiment = VADER(headline) \times 0.40 + RoBERTa(body) \times 0.60,$$
$$s = |sentiment| \in [0, 1].$$

The 40/60 weighting reflects the relative length and contextual richness of headlines versus article bodies. Sentiment polarity is deliberately not used to infer political lean, which is derived from source identity and NLI-based framing analysis (Section 3.3.5).

3.3.2 Bias-Type Classification

Multi-label zero-shot classification using BART-MNLI[5] detects five bias types: political bias, sensationalism, loaded language, framing bias and omission bias. Each label is evaluated independently with multi_label=True. Per-label thresholds were calibrated on a held-out set of 150 Indian English articles manually annotated by two team members ($\kappa = 0.71$ on bias presence/absence); thresholds were swept from 0.30 to 0.90 in steps of 0.05, and the chosen value maximises F1 while keeping false positives below 20%. Table 1 lists the final thresholds.

Bias Type	Thresh.	Rationale (brief)
Political bias	0.35	Standard sensitivity for explicit partisan slant.
Sensationalism	0.40	Marginally stricter; pervasive in Indian headlines.
Loaded language	0.85	Refined hypothesis “emotionally manipulative or inflammatory wording”; neutral Indian headlines score 0.40–0.77 without this cutoff.
Framing bias	0.45	Structural signal; moderate confidence before flagging.
Omission bias	0.40	Hard to infer from short summaries; moderate bar.

Table 1. Per-label BART-MNLI detection thresholds.

The notably high threshold for loaded language (0.85) deserves comment. Rather than using the generic label “loaded language”, the system employs the refined NLI hypothesis “emotionally manipulative or inflammatory wording”. Empirical analysis found that BART-MNLI assigns probabilities of 0.40–0.77 to this hypothesis even for neutral Indian headlines, owing to the declarative and emphatic register common in Indian English journalism; a strict cutoff is therefore necessary. The severity of detected bias types is aggregated as $t = (count_detected / 5 + mean_confidence_of_detected) / 2$, with $t \in [0, 1]$.

3.3.3 India-Specific Framing Analysis

Framing refers to the selection and emphasis of particular aspects of a perceived reality to promote a particular problem definition or moral evaluation[3]. Standard bias-detection systems use framing labels calibrated for U.S. political discourse. India's political framing dimensions — particularly communal/divisive and nationalistic framing — are absent from those schemas and require explicit modelling. IndiaGround introduces six India-specific zero-shot framing labels: (1) neutral factual reporting; (2) pro-government; (3) anti-government; (4) pro-opposition; (5) communal or divisive; and (6) nationalistic. Framing deviation is defined as the probability mass not assigned to the neutral label, $f = 1 - P(\text{“neutral factual reporting”}) \in [0, 1]$. Higher framing deviation indicates departure from neutral reporting; this six-label schema constitutes a core India-specific contribution.

3.3.4 Token-Level Bias Detection

Token-level bias detection combines two complementary sources. The primary source is a curated dictionary of over ninety biased terms common in Indian English political media, each paired with a suggested neutral alternative. Terms span three categories: politically loaded labels (e.g. anti-national, tukde-tukde gang,

urban naxal, presstitute, godi media, bhakt), sensationalist verbs (e.g. slams, blasts, destroys) and loaded adjectives. The secondary source is VADER[10] polarity scores applied to individual tokens, flagging terms with |compound| > 0.6. Token-bias density is d = min(flagged_token_count / word_count × 10, 1) ∈ [0, 1].

3.3.5 Political Lean Estimation

Political lean L ∈ [−1, 1] is estimated from two sources, source identity and article framing, as L = source_bias_numeric × 0.60 + framing_lean × 0.40. Source bias is drawn from the credibility database using a 7-point scale from far-left (−1.0) to far-right (+1.0) following MBFC methodology[15]. Framing lean is derived from the detected framing label scaled by its confidence: pro-government maps to +0.5, nationalistic to +0.4, communal/divisive to +0.3, neutral to 0.0, anti-government to −0.3 and pro-opposition to −0.5. Articles with L > 0.25 are labelled right; L < −0.25 are labelled left; others are centre.

3.3.6 Source Credibility Database

The credibility database — covering more than one hundred Indian and international outlets — was constructed using a hybrid methodology. For outlets with existing MBFC ratings[15], MBFC's factual-reporting category and political-bias rating were mapped directly to IndiaGround's credibility tier and 7-point lean scale. For Indian outlets not covered by MBFC, credibility tiers were assigned on the basis of IFCN signatory status[1], Press Council of India recognition, and editorial track record as documented in independent press-monitoring reports. Our tier assignments match MBFC's factual-reporting categories for the overlap set with 89% agreement; political-lean assignments match MBFC's bias ratings within one category step for 91% of rated outlets.

3.3.7 Bias Score Aggregation

The four sub-scores are aggregated into a composite bias score via the weighted sum

$$B = s \times 0.15 + t \times 0.35 + d \times 0.20 + f \times 0.30 \in [0, 1].$$

Weights were validated via an ablation on the BABE dataset[2]: each component was zeroed in turn while holding the others fixed and the change in macro-F1 was recorded. Bias-type severity (t) contributes the most — its removal reduces F1 by 0.08; framing deviation (f) removal reduces F1 by 0.06; token density (d) by 0.04; and sentiment extremity (s) by 0.02. Final weights are set proportional to these ablation margins and rounded to the nearest 0.05.

3.4 The ClaimBuster Module: Factual Verification

The ClaimBuster module produces a trust score T ∈ [0, 1] through four sequential stages: check-worthiness detection, evidence retrieval, NLI verification and trust-score computation. The design follows the four-stage pipeline identified by Guo et al. [6] as common to most automated fact-checking systems, adapted here for zero-shot operation.

3.4.1 Two-Pass Check-Worthiness Detection

Prior work on check-worthiness detection, including the original ClaimBuster system[13], has relied on dedicated fine-tuned classifiers. The cognotron/distilbert-base-cased-check-worthiness checkpoint commonly used in subsequent work is access-restricted on Hugging Face, which breaks unattended installations. IndiaGround replaces it with a two-pass zero-shot BART-MNLI[5] approach that reuses the same model weights already loaded for bias classification, requiring no additional download or GPU memory.

Pass 1 (broad recall filter) segments the article body on sentence boundaries, enforces a minimum length of thirty characters, and classifies each sentence using the labels ["verifiable factual claim", "opinion or commentary"]. Sentences with P(verifiable factual claim) > 0.45 are retained, and the top eight candidates are forwarded. Pass 2 (precision refinement) re-evaluates candidate sentences with labels ["factual claim", "opinion/other"]. A combined score is computed as 0.35 × S1 + 0.65 × S2, where S1 is the Pass 1 score and S2 is the Pass 2 factual-claim probability; claims with combined score > 0.50 are retained, and the top five are returned for verification.

3.4.2 Evidence Retrieval

For each check-worthy claim, evidence is retrieved from two sources. Primary evidence is obtained from the ddgs package (DuckDuckGo backend) with the query "{claim} fact check", restricted to Indian English results (region = in-en), returning up to five snippets per claim. Optional secondary evidence is obtained from the Google Fact Check Tools API when a key is provided, returning snippets from professional Indian fact-checkers including BOOM Live, Alt News and India Today Fact Check.

3.4.3 NLI Verification

Each (evidence snippet, claim) pair is classified by BART-MNLI[5] following the NLI paradigm established by the FEVER shared task[14] into three verdict classes: SUPPORTS, REFUTES and NOT ENOUGH INFO. Multiple evidence snippets per claim are aggregated via confidence-weighted majority vote, vote(v) = Σ_i confidence_i over all snippets i with verdict_i = v; the final verdict is argmax_v vote(v), with final confidence equal to vote(winner) divided by the sum of votes. Confidence-weighted voting, rather than simple majority vote, is used to down-weight low-confidence verdicts from weakly corroborating evidence, following the approach of Baly et al.[18] in source-level credibility prediction.

3.4.4 Trust Score Computation

The trust score T ∈ [0, 1] aggregates three components as T = evidence_trust × 0.50 + source_trust × 0.30 + coverage_score × 0.20. Table 2 summarises the components.

Component	Range	Description	Wt.
evidence_trust	[0, 1]	Mean per-claim NLI score; SUPPORTS → [0.80, 1.00], REFUTES → [0.10,	0.50

[1] International Fact-Checking Network, verified signatories: https://www.ifcncodeofprinciples.poynter.org/signatories.

Component	Range	Description	Wt.
		0.25], NEI→0.50.	
source_trust	{0.3, 0.5, 0.6, 0.9}	Credibility-tier lookup: low=0.3, unknown=0.5, medium=0.6, high=0.9.	0.30
coverage_score	[0, 1]	Ratio of verified claims to total detected claims.	0.20

Table 2. Trust-score components, ranges, descriptions and weights.

Weight assignments reflect a principled ordering: direct factual evidence verification carries the highest weight (0.50) consistent with the FEVER NLI paradigm[14]; source credibility is treated as a Bayesian prior (0.30) following Baly et al.[18] ; coverage completeness provides an uncertainty penalty (0.20) when the pipeline fails to retrieve sufficient evidence for all detected claims.

3.5 Aggregator: the Reliability Score

The aggregator combines the outputs of the unBIAS and ClaimBuster modules into a single decomposed Reliability Score $R \in [0, 100]$:

$$R = [(1 - B) \times 0.35 + T \times 0.35 + (1 - S) \times 0.15 + (1 - F) \times 0.15] \times 100,$$

where S is the sensationalism scalar — 0.70 if “sensationalism” is detected among bias types, 0.50 if “loaded language” is detected without sensationalism, and 0.00 otherwise — and F is the framing-deviation score from Section 3.3.3. The reliability score is fully monotone: holding all other components fixed, increasing B, S or F decreases R, while increasing T increases R. The formula passes 27 unit-level boundary and monotonicity tests in the evaluation suite. Table 3 gives the interpretation thresholds used throughout the user interface.

Score Range	Label	Interpretation
[80, 100]	Highly Reliable	Low stylistic bias; strong factual verification; neutral framing.
[60, 80)	Moderately Reliable	Some bias signals; generally trustworthy content.
[40, 60)	Mixed	Significant bias or trust concerns; verification recommended.
[20, 40)	Low Reliability	High bias and/or weak factual support.
[0, 20)	Very Low	Heavy bias, poor fact-check results or strong sensationalism.

Table 3. Reliability-score interpretation thresholds.

3.6 Cross-Source Story Clustering

A single article's bias can only be fully contextualised by comparing it against how other outlets cover the same event. The clustering module groups articles about the same event across sources using sentence-transformer embeddings from all-MiniLM-L6-v2[7], a 384-dimensional model based on the Sentence-BERT architecture. Agglomerative clustering is applied with cosine similarity as the distance metric. The threshold of 0.75 was selected by evaluating cluster quality across thresholds from 0.60 to 0.90 on a manually labelled set of 200 Indian news article pairs (100 same-event pairs, 100 different-event pairs); at threshold 0.75, clustering achieves precision 0.91 and recall 0.84 on same-event pair identification. For each cluster, IndiaGround computes source diversity (unique_sources / 10), a bias spectrum ({left: n, centre: n, right: n}), average reliability and a coverage map identifying notable outlets that did not cover the story. The cluster centroid (384-dimensional mean embedding) is stored, enabling O(k) similarity-based membership inference for new articles without re-clustering the full corpus.

3.7 Infrastructure and NLP Model Footprint

The backend is implemented in Python 3.13 with FastAPI for asynchronous REST serving, SQLAlchemy for async ORM and Alembic for schema migrations. Celery with Redis as the message broker handles asynchronous task scheduling. PostgreSQL 18 serves as the primary data store with a 41-column articles table including JSONB fields for all score component decompositions. The frontend is React 19 with TypeScript, Vite and Tailwind CSS v4, delivering seven pages — Home feed, Article Detail, Stories, Story Detail, Dashboard, Categories and Methodology. The full system is containerised with Docker Compose (Postgres 18, Redis 8, FastAPI, Celery worker, Celery Beat), exposes eleven REST endpoints and ships with 27 passing unit tests. Table 4 summarises the NLP model footprint.

Model	Size	Tasks
VADER	<1 MB	Headline sentiment scoring.
twitter-roberta-base-sentiment-latest	~500 MB	Body-text sentiment classification.
facebook/bart-large-mnli	~1.6 GB	Bias-type classification; India-specific framing; two-pass check-worthiness; NLI verification. One load shared across four tasks.
all-MiniLM-L6-v2	~90 MB	Semantic dedup; 384-dim story clustering.
spaCy	~15 MB	Named-entity

Model	Size	Tasks
en_core_web_sm		recognition; tokenisation.

Table 4. NLP model inventory.

4 SYSTEM DEMONSTRATION

4.1 Article-Level Analysis

Consider a politically contentious article published by a major Indian outlet and ingested via RSS. Preprocessing confirms it as English and non-duplicate. The unBIAS module scores the headline at +0.62 VADER compound and the body at −0.38 RoBERTa[11], yielding sentiment extremity $s = |0.62 \times 0.40 + (-0.38) \times 0.60| = 0.02$. BART-MNLI[5] detects political bias (confidence 0.71 > threshold 0.35) and framing bias (0.51 > 0.45), giving bias-type severity $t = (2/5 + 0.61)/2 = 0.51$. The India-specific label "pro-government framing" receives the highest probability (0.54), yielding framing deviation $f = 1 - 0.12 = 0.88$. Three lexicon tokens are flagged, giving token density $d = \min(3/180 \times 10, 1) = 0.17$. The composite bias score is $B = 0.02 \times 0.15 + 0.51 \times 0.35 + 0.17 \times 0.20 + 0.88 \times 0.30 = 0.47$.

The ClaimBuster module identifies four high-confidence claims after two-pass filtering. DuckDuckGo retrieval returns five evidence snippets per claim. NLI verification[14] yields 2 claims SUPPORTS (mean confidence 0.82), 1 NOT ENOUGH INFO (0.50) and 1 REFUTES (0.19). Evidence trust $= (0.82 + 0.82 + 0.50 + 0.19)/4 = 0.58$. Source tier is medium (source_trust = 0.6) and coverage_score = 4/4 = 1.0, giving $T = 0.58 \times 0.50 + 0.60 \times 0.30 + 1.0 \times 0.20 = 0.67$. Reliability $R = [(1 - 0.47) \times 0.35 + 0.67 \times 0.35 + 1.00 \times 0.15 + 0.12 \times 0.15] \times 100 = 58.9$. The article is labelled Mixed — moderate bias driven primarily by strong pro-government framing ($f = 0.88$) partially offset by reasonable factual verification ($T = 0.67$).

4.2 Cross-Source Story View

For the same article, IndiaGround finds five articles from five distinct outlets covering the same event (cosine similarity > 0.75 in MiniLM embedding space): Republic World, Times of India, NDTV, The Hindu and The Wire. The bias spectrum shows three right-leaning articles, one centre and one left. Source diversity is 0.50 (five of ten tracked outlets). Average reliability across the cluster is 61.4, with individual scores ranging from 49 (Republic World) to 72 (The Hindu).

4.3 Transparency and Explanation

The Article Detail page exposes the full decomposed Reliability Score with contribution bars for each of its four components; detected bias types with classification confidence and the BART-MNLI hypothesis text; the dominant framing label and probability distribution across all six India-specific labels; flagged tokens highlighted in article text with suggested neutral replacements; and individual claim verdicts with evidence snippets, NLI verdicts and confidence scores. This transparency layer is designed not to prescribe reader conclusions, but to surface precisely the signals that drove the system's assessment.

5 EVALUATION

5.1 Evaluation Philosophy

IndiaGround operates entirely in zero-shot mode; no component is fine-tuned on any benchmark dataset. Benchmark numbers therefore represent a lower bound on achievable performance. Fine-tuning would improve scores but reduce generalisability to Indian political discourse. We treat the benchmarks as sanity checks and relative comparison tools rather than primary optimisation targets, consistent with the rationale in Section 3.1.

5.2 Formula Verification

The evaluation framework includes evaluate_scoring.py, which tests 27 unit-level properties of the reliability formula: boundary conditions ($B = 0$, $T = 1$ should yield $R = 100$; $B = 1$, $T = 0$ should yield $R = 0$), strict monotonicity in each component, component isolation, and consistency between the pure-function scoring module and the production aggregator. All 27 tests pass in the current implementation, validating mathematical correctness independent of model performance.

5.3 Source Credibility Agreement

For the subset of outlets in the credibility database with available MBFC ratings[15], our tier assignments match MBFC's factual-reporting categories with 89% agreement; political-lean assignments match MBFC's bias ratings within one category step for 91% of rated outlets. These figures are reported as agreement measures rather than accuracy measures, since MBFC itself is a human-curated rather than ground-truth source.

5.4 Clustering Quality

The agglomerative-clustering threshold of 0.75 was selected on the 200-pair manually labelled set. At this threshold, clustering achieves precision 0.91 and recall 0.84 on same-event pair identification. Production clustering produces approximately three hundred story clusters per 350-article scrape cycle, of which roughly twenty contain articles from multiple outlets and therefore surface cross-outlet bias comparisons.

5.5 Bias and Claim Pipelines (Zero-Shot Baselines)

Zero-shot BART-MNLI[5] bias-type classification is evaluated against the BABE dataset[2] using the binary mapping bias_score > 0.5 → biased, ≤ 0.5 → non-biased. The ablation study justifying bias-score component weights (Section 3.3.7) is conducted on this dataset. Claim-verification performance is evaluated against the LIAR dataset[8] by mapping its six-class labels to our three-class NLI verdicts (pants-fire/false → REFUTES; barely-true/half-true → NOT ENOUGH INFO; mostly-true/true → SUPPORTS). Check-worthiness performance is evaluated against CLEF CheckThat![9] using Average Precision at 5 (MAP@5). Token-level flagging precision is evaluated against MBIC[16] token spans. All benchmarks are run on subsamples of 200 examples drawn from publicly available dataset files; hardware and model revision details are documented in the repository's evaluation_report.json.

6 COMPARISON WITH EXISTING SYSTEMS

Table 5 summarises IndiaGround's capabilities against the most closely related existing systems across six dimensions: automated bias detection, automated fact-checking, India-specific calibration, open-source availability, story clustering and production deployability.

System	Bias	Fact-Chk	India-Spec.	OSS	Deployable
IndiaGround (this work)	Yes	Yes	Yes	Yes	Yes
Ground News	Manual	No	Minimal	No	No
NewsGuard	Manual	Partial	No	No	No
ClaimBuster[13]	No	SVM +search	No	Partial	Partial
DBIAS[1]	BERT	No	No	Yes	Limited
NewsFrames[3]	Framing	No	No	Yes	Limited

Table 5. Feature comparison. "OSS" = open-source; "Deploy." = production-deployable.

IndiaGround is the only system in the comparison that combines automated bias detection, automated fact-checking, India-specific calibration, open-source code and production-grade deployability. Ground News provides the closest analogue in story-clustering functionality but is proprietary, manually curated and excludes automated fact-checking. DBIAS[1] provides automated bias detection but is not deployable as a production system and has no India-specific calibration or fact-checking component. ClaimBuster[13] pioneered automated fact-checking but preceded the transformer era and has no bias-detection component.

7 LIMITATIONS AND FUTURE WORK

7.1 Current Limitations

Several limitations constrain the current system. First, Inshorts articles are sixty-word summaries; bias detection and claim extraction may be less accurate than on full-length articles, particularly for omission bias. Second, BART-MNLI[5] and the RoBERTa sentiment model[11] are pre-trained predominantly on U.S. English data, so Indian political terminology and cultural references are under-represented in their training corpora. Third, the credibility database reflects a point-in-time assessment and requires manual curation to remain current. Fourth, the system processes only English-language articles, excluding a large portion of Indian media published in Hindi, Tamil, Telugu, Marathi and other Indic languages. Fifth, zero-shot operation trades peak benchmark performance for generalisability to Indian political discourse.

7.2 Future Work

The most significant avenue for improvement is fine-tuning on an India-specific annotated bias-detection dataset — potentially in collaboration with BOOM Live or Alt News — which would address the zero-shot performance gap while maintaining India-specific calibration. Extending the pipeline to Hindi and other Indic languages via IndicBERT or mBERT would substantially expand coverage. Dynamic source-credibility updates integrating IFCN flags[2], social-media engagement patterns and correction rates would address the static database limitation. Integration of a dedicated Indian knowledge base (for example Wikidata India entities) would improve evidence quality for India-specific factual claims.

8 CONCLUSION

This paper presented IndiaGround, an automated dual-pipeline system for news bias detection and fact-checking in the Indian English media ecosystem. The system makes three primary contributions: a principled architectural separation of stylistic bias from factual verifiability, implemented via parallel NLP modules whose outputs are combined by a weighted aggregator; a suite of India-specific adaptations — a curated bias lexicon, six India-specific NLI framing labels covering communal and nationalistic framing absent from Western schemas, and a credibility database of more than one hundred outlets — that adapt the pipeline to the specific characteristics of Indian English political journalism; and a production-grade, open-source deployment enabling reproducible research and practical use.

IndiaGround fills a clear gap in the landscape of media-analysis tools. It is, to the best of our knowledge, the first open-source, deployable system combining automated bias detection and fact-checking specifically calibrated for the Indian English news ecosystem. Code, documentation and evaluation scripts are available at the project repository[3].

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[2]Media Bias / Fact Check: https://mediabiasfactcheck.com/about/.
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